

Echoes of the Matchlock: Playing Shooting Games Using Cultural Heritage in Mixed Reality

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Figure 1: 3D model of the matchlock gun with major component labels.

Abstract

Echoes of the Matchlock introduces an innovative Mixed Reality (MR) system with entertainment elements that transforms cultural heritage education by making Japanese historical matchlock firearms engaging for younger audiences. Traditional museum displays often fail to capture young people's interest in historical artifacts, creating a gap in cultural knowledge transmission. This system addresses this challenge by integrating entertaining target shooting experiences with educational content about matchlock mechanisms. Users handle a physical 3D-printed matchlock stock replica embedded with Meta Quest 3 controllers while engaging in shooting activities that reveal the weapon's internal firing process through MR visualization. The system allows component disassembly and name display for deeper understanding. By combining entertainment features with cultural education, this approach successfully engages diverse age groups while preserving and transmitting traditional craftsmanship knowledge.

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1 Introduction

Cultural heritage preservation faces a significant challenge in the digital age: how to engage younger audiences who have grown up with interactive digital media and gaming experiences. Traditional museum displays and educational materials, while informative, often fail to capture the attention and interest of digital natives

who expect dynamic, interactive, and entertaining content. This generational gap threatens the transmission of valuable historical knowledge and appreciation for traditional craftsmanship.

The Japanese matchlock gun (hinawaju) represents a particularly compelling case study for this challenge. While the matchlock is widely recognized throughout Japan due to its pivotal role in the country's history since its introduction by Portuguese traders at Tanegashima in the 16th century [2003], most people—not only young people but also many non-specialists—have limited opportunities to understand its internal structure and operational principles. The matchlock's construction and firing mechanism are actually quite simple in concept; however, existing educational materials struggle to convey the operational details intuitively because static images and textual descriptions are inherently limited media for demonstrating three-dimensional mechanical processes. The weapon's firing mechanism involves the coordinated movement of multiple components (Figure 1) that remain difficult to comprehend without dynamic visual demonstration.

Previous research has addressed this educational challenge through various digital approaches. Hayashi et al. developed 3D computer graphics animations and augmented reality (AR) content to visualize matchlock construction and firing mechanisms [2023; 2025; 2024]. While these efforts successfully demonstrated the technical feasibility of digital cultural heritage education, they primarily focused on conveying information rather than creating engaging experiences.

This paper presents "Echoes of the Matchlock," an innovative Mixed Reality (MR) system that reimagines cultural heritage education through entertainment-driven interaction. Rather than simply digitizing traditional educational content, our approach transforms learning into an engaging shooting experience that naturally reveals the matchlock's operational principles. The system integrates a physical 3D-printed matchlock stock replica embedded with Meta Quest 3 controllers, allowing users to handle a tangible artifact while experiencing immersive MR visualizations. This paper focuses on the features designed to help users understand the matchlock's structure.

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The system is expected to address the fundamental challenge of making historical artifacts relevant and engaging to contemporary audiences, particularly those who might otherwise show little interest in traditional cultural heritage. By embedding educational content within entertaining shooting activities, users are anticipated to naturally discover the matchlock's internal mechanisms while pursuing game objectives, creating what we term "stealth learning."

2 User Interaction

In this program, we utilized the pass-through functionality of the Meta Quest 3 for development, allowing users to see their surrounding environment while wearing the headset. Under these conditions, we created a 3D-printed matchlock stock model that precisely embeds the Meta Quest 3 right-hand controller. The stock's grip shape was modeled based on laser-scanned data from an authentic matchlock gun, providing a realistic tactile experience that closely mimics handling the original artifact.

The physical stock is rigidly attached to the controller. Position matching is maintained through Unity's transform hierarchy, where the gun model is parented to the controller's tracked pose, while individual virtual components utilize rotational interpolation for smooth visual response synchronized with user input. This enables users to experience holding the CG model of the matchlock while the physical replica provides tactile feedback, as shown in Figure 2 (a).

However, the embedding of the controller results in many controller buttons being occluded, as illustrated in Figure 2 (b). This constraint limited our user interface design to only the joystick and trigger buttons. Ultimately, we decided to implement four interaction modes: joystick up movement, joystick down movement, vertical joystick press, and trigger press.

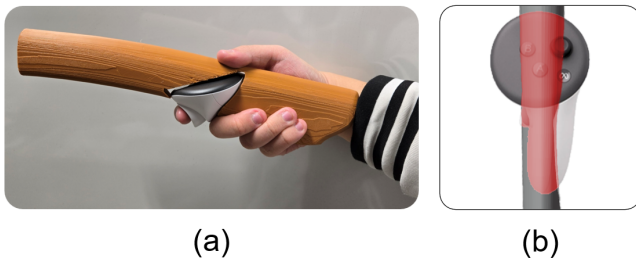


Figure 2: User interaction. (a) The matchlock stock replica with Meta Quest 3 held by the user. (b) Red areas indicate buttons that are occluded and unusable due to the replica model.

3 Understanding the Matchlock Structure Intuitively

The matchlock gun is a firearm approximately 123 centimeters in length, composed of various components including but not limited to those shown in Figure 1. In our program, users can disassemble or reassemble the matchlock components by moving the joystick up and down. These components slowly move to their expanded

or assembled positions following the user's joystick operations. During this process, users can vertically press the controller's joystick button to show or hide the names of these components. For example, Figure 3 (a) shows the fully disassembled gun body with component names displayed.

When users fire the weapon, three firing-related components (Trigger, Sear, Serpentine lock) gradually change their positions according to the trigger button's depression level, as shown in Figure 3 (b), allowing users to intuitively understand the matchlock's structure during firing. Finally, when the trigger is fully depressed, a bullet is fired from the gun accompanied by a loud sound effect, establishing an important functional foundation for the target shooting game.

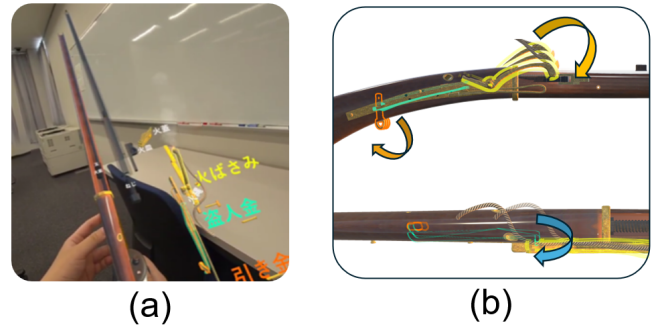


Figure 3: Matchlock gun structure. (a) Disassembled matchlock model displaying the names of key components. (b) Components that move when the trigger is pressed (Trigger, Sear, Serpentine lock), with side view above and top view below.

4 Conclusion

This paper presented "Echoes of the Matchlock," an MR system integrating physical replicas with virtual visualization to teach matchlock mechanisms through entertaining shooting activities.

The system has received positive user feedback. Analysis of these responses represents our next work.

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